OPPORTUNITY BRIEF · SALES AND PRESALES

Deploy ML-based anomaly detection on multivariate sensor time-series to predict equipment failures and reduce unplanned downtime in discrete manufacturing plants.

Manufacturing × Predictive maintenance × Machine learning · OS015 v3 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
24 is the world moving	**85** can we play, can we win	**MEDIUM** 0 named in the evidence	**€309m** observed evidence · per year	**LATER** research and standards activi...	**L0** 18 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

We can help discrete manufacturers predict equipment failures before they happen by deploying machine learning on their existing sensor data, reducing unplanned downtime. This opportunity exists now because recent advances in federated learning, conformal prediction, and synthetic data generation have made anomaly detection more reliable and privacy-preserving, even with limited labeled failure data.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€431m** €77.6m – €2.3bn	**€309m** €55.7m – €1.7bn	**€18.6m** €3.3m – €99.8m	observed
Observed: tendered contract value, annualised	**€20.9m** €20.9m – €56.6m	**€20.9m** €20.9m – €56.6m	**€1.3m** €1.3m – €3.4m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Machine learning	3.3 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TML	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 3 of 3 declared geographies are outside the European reference data (CN, SA, CA); the estimate covers the European footprint only.

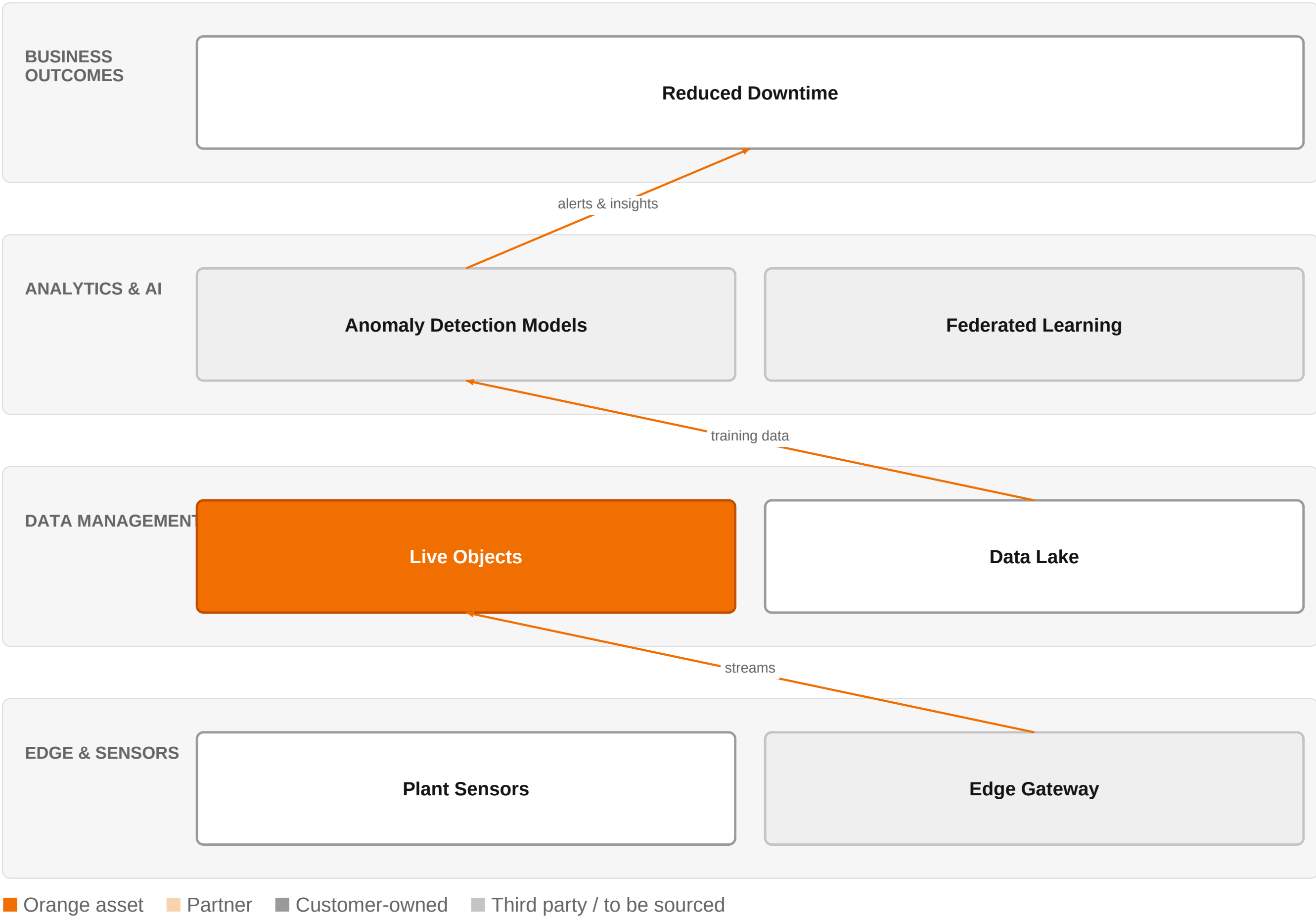
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 3 of 3 declared geographies are outside the European reference data (CN, SA, CA); the estimate covers the European footprint only.

The solution, in one picture

Predictive Maintenance Solution Architecture



This architecture shows how Orange's IoT and AI capabilities combine with customer data to deliver predictive maintenance insights.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Recent research shows that federated learning enables collaborative anomaly detection across distributed industrial IoT without sharing raw data, addressing privacy and security concerns [SIG-C6ED90595BBC, SIG-F633044AF87A]. Conformal machine learning provides reliable anomaly detection with statistical guarantees for industrial cyber-physical systems [SIG-89E9D386676C]. Synthetic data generation with deep generative models helps overcome the small-sample problem in equipment anomaly detection [SIG-3B2336BA027B]. These advances address long-standing challenges of data scarcity, privacy, and reliability in industrial time-series anomaly detection.

`Sources: SIG-C6ED90595BBC openalex.org 2026-04-18; SIG-F633044AF87A openalex.org 2026-01-29; SIG-89E9D386676C openalex.org 2026-02-14; SIG-3B2336BA027B openalex.org 2026-02-23`

Who buys, and why now

The buyer is typically the VP of Manufacturing or Head of Plant Operations, who owns the plant's uptime and maintenance budget. The pain is felt by maintenance engineers who are firefighting unplanned breakdowns, and by production managers who lose output when a line stops. The trigger is often a recent high-impact failure, a new corporate initiative to reduce downtime, or a push from Industry 4.0 programs to leverage existing sensor data.

Why it is hot now — every claim bound to a dated source

- Collaborative and privacy-preserving federated learning is used for industrial time-series anomaly detection. [SIG-C6ED90595BBC]
- A secure federated learning framework for anomaly detection in smart industrial networks integrates adaptive anomaly detection and privacy preservation. [SIG-F633044AF87A]

- Federated learning enables collaborative anomaly detection across distributed industrial IoT without sharing raw data, addressing privacy and security concerns. [SIG-C6ED90595BBC, SIG-F633044AF87A]
- Federated learning is applied to secure industrial automation and grid optimization, covering predictive maintenance and fault detection. [SIG-C201BF32AE96]
- Anomaly detection in industrial time-series is critical for equipment health monitoring and predictive maintenance. [SIG-8EE4F2310EA0, SIG-EB5664191E1A]
- Conformal machine learning provides reliable anomaly detection with statistical guarantees for industrial cyber-physical systems. [SIG-89E9D386676C]
- Advanced Metering Infrastructure provides high-resolution electrical data for enhancing industrial efficiency and monitoring equipment health, but anomalies compromise the data's utility. [SIG-8EE4F2310EA0]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a team of AI, Data, and Cloud experts and IoT experts to design and deploy a predictive maintenance solution. The engagement would start with a discovery workshop to understand the customer's sensor infrastructure and pain points, then leverage Live Objects to securely collect and manage IoT data. The team would build and train anomaly detection models, possibly using federated learning to respect data privacy, and integrate with the customer's existing systems. Certifications like ISO 27001 and TISAX would be applied to ensure security and compliance, especially for automotive customers.

Why Orange can win

Orange brings a unique combination of IoT expertise (Live Objects), AI/Data/Cloud experts, and strong security certifications (ISO 27001, TISAX) that are critical for manufacturing customers. The ability to leverage Orange Group researchers for cutting-edge techniques like federated learning and conformal prediction gives us an edge. However, the asset list is thin on specific manufacturing references, so we must rely on our partner ecosystem (AWS, Google Cloud, NVIDIA) to fill gaps.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NiCE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Machine learning; competes in OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning	structural	—
NiCE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning	structural	—
Dataiku	Data and AI platform	sells Machine learning	structural	—
Palantir	Data and AI platform	sells Machine learning; competes in OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. What is your current unplanned downtime rate and what is the biggest cause?
2. Do you have a centralized data lake or are your sensor data siloed across plants?
3. Have you tried any predictive maintenance pilots before? What happened?
4. What is your data governance stance on sharing sensor data across plants or with third parties?
5. Which equipment types cause the most costly failures?
6. What is your IT/OT security certification requirement for new solutions?

Objections you will meet

They will say	You can answer
We already have a predictive maintenance solution from a hyperscaler.	That's true, and hyperscalers have strong platforms. But our advantage is in integrating across your entire IT/OT landscape and bringing in security certifications like ISO 27001 and TISAX that are critical for manufacturing. We can also leverage our partner ecosystem to complement what you have.
Our data is too sensitive to share with an external provider.	We understand data sensitivity is a top concern. That's why we use federated learning techniques that keep data on your premises, and we can deploy our solution within your security perimeter. Our ISO 27001 certification and TISAX for automotive ensure compliance.
We don't have enough labeled failure data to train a model.	That's a common challenge. Recent advances in synthetic data generation and unsupervised learning can help. We can also start with a small pilot on one line to demonstrate value before scaling.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-C6ED90595BBC	2026-04-18	openalex.org	1	Towards collaborative and privacy-preserving industrial time-series anomaly detection with fede...
SIG-3B2336BA027B	2026-02-23	openalex.org	1	SyntheITS: Synthetic industrial time-series data with prior knowledge and deep generative model...
SIG-89E9D386676C	2026-02-14	openalex.org	1	Conformal machine learning for reliable anomaly detection in industrial cyber-physical systems
SIG-03DDF2041F65	2026-02-03	openalex.org	1	Anomaly detection for industrial time series in process industry using informed machine learnin...
SIG-8EE4F2310EA0	2026-02-01	openalex.org	1	A Comparative Analysis of Machine Learning Models for Anomaly Detection in Industrial Smart Met...
SIG-EB5664191E1A	2026-02-01	openalex.org	1	A Systematic Review of Anomaly and Fault Detection Using Machine Learning for Industrial Machin...
SIG-F633044AF87A	2026-01-29	openalex.org	1	SecuFL-IoT: an adaptive privacy-preserving federated learning framework for anomaly detection i...
SIG-C201BF32AE96	2026-01-07	openalex.org	1	Federated Learning for Secure Industrial Automation and Grid Optimization
SIG-A98EBAF4B99E	2026-01-01	openalex.org	1	Adaptive Latent Distribution Modeling for Industrial Time Series Anomaly Detection
SIG-0E742BED215C	2025-12-23	openalex.org	1	Improving Multivariate Time-Series Anomaly Detection in Industrial Sensor Networks Using Entrop...
SIG-1C6BA6D836AF	2025-12-17	openalex.org	1	FFT-Guided Multi-Window USAD with DTW–Isolation Forest for Reliable Anomaly Detection in Indust...
SIG-A320B94D827D	2025-12-06	openalex.org	1	Anomaly detection model based on unsupervised learning for multivariate industrial time series
SIG-1157B261CA95	2025-11-27	openalex.org	1	A Reliable Framework for Human-in-the-Loop Anomaly Detection in Time Series
SIG-76E3106EFB64	2025-10-18	openalex.org	1	A structure-aware routing based anomaly detection for industrial multi-sensor time series
SIG-174AD17494CC	2025-09-12	openalex.org	1	Transformer-based Sequential Contrastive Learning for Industrial Robot Time Series Anomaly Dete...
SIG-81F8AA75428C	2025-09-09	openalex.org	1	Time-Series-Based Anomaly Detection in Industrial Control Systems Using Generative Adversarial ...
SIG-0F24EBB350CC	2025-09-08	openalex.org	1	Multivariate Time Series Anomaly Detection Based on Inverted Transformer with Multivariate Memo...
SIG-5C50B186142A	2026-04-30	arxiv.org	3	PhaseNet++: Phase-Aware Frequency-Domain Anomaly Detection for Industrial Control Systems via P...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); risks_and_unknowns (missing from the model output); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:00+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.